

High-frequency vibrational power flows in randomly heterogeneous coupled structures

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Abstract The purpose of this paper is to expound some recent developments for the modeling and numerical simulation of high-frequency (HF) vibrations of randomly heterogeneous structures, and outline some perspectives of future research. The mathematical-mechanical model is based on a microlocal analysis of quantum or classical linear wave systems. The theory shows that the energy density associated to their mean-zero solutions—including the strongly oscillating (HF) ones—satisfies a Liouville-type transport equation, or a radiative transfer equation in a random medium at length scales comparable to the small wavelength. Its main limitation to date lies in the consideration of energetic boundary and interface conditions consistent with the boundary and interface conditions imposed to the solutions of the wave system. The corresponding power flow reflection/transmission operators are derived formally and rigorously for elastic media, including slender structures, near the doubly-hyperbolic and hyperbolic-elliptic sets, ignoring however the glancing set. Yet a radiative transfer model in bounded media with general transverse or diffuse boundary conditions for the power flows is detailed in the paper. Some direct numerical simulations are presented to illustrate the theory.

1 Introduction

Engineering structures exhibit typical transport and diffusive behaviors in the higher frequency range of vibration. These regimes are described by linear transport equations for the energy density associated to the oscillating solutions of the Navier-Cauchy equation for elastic wave propagation. More generally this result holds for all symmetric hyperbolic systems, including quantum waves and classical acoustic or electromagnetic waves [1, 2, 3, 4, 5, 6, 7]. It has been specialized to slender elastic

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structures, typically beams, plates and shells, and fluid-saturated poro-visco-elastic media in [8, 9, 10, 11] for applications to aerospace structures. The main limitation to date of the theory lies in the consideration of boundary and interface conditions for quadratic quantities (e.g., the energy and power flow densities) consistent with the boundary and interface conditions imposed to the displacement and stress fields. This issue is dealt with in [12] starting from an approximate high-frequency solution of the Navier-Cauchy equation for which an explicit formula of the associated energy density can be derived, the latter being also the energy density associated to the true solution, in the L^2 -norm sense. This approximate solution is chosen for example as the superposition of Gaussian beams weighted by the FBI transform of the initial conditions away from the boundaries. Yet it is always possible to construct Gaussian beams which satisfy Dirichlet or Neumann boundary conditions, and their energy densities are transported along the bicharacteristic lines of the underlying wave equation. Thus the reflections of energy flows are described in phase space by those of the Gaussian beams. Notably the physical space subset yielding transverse rays is transported along the bicharacteristic lines reflected on the boundaries according to Descartes law (specular reflections). Based on these results, a formal derivation of high-frequency power flow reflection/transmission coefficients at interfaces between slender sub-structures such as beams or plates has been proposed in [11]. More recently, a systematic derivation of the elastic energy density boundary conditions for Dirichlet or Neumann boundary conditions imposed on the displacement or stress fields near the doubly-hyperbolic set (where both types of elastic waves are incident on the boundary) and the hyperbolic-elliptic set (where transverse elastic waves are incident on the boundary with an angle greater than their critical angle) has been proposed in [13]. The analysis however does not account for the glancing set and the transport of energy along an interface by so-called gliding rays. The proofs follow the original ideas of Miller [14], who in return also considered the glancing set for *scalar* waves. The analysis of surface effects (glancing) in the elastic case for *polarized* waves is the subject of ongoing research.

The purpose of this paper is to expound a transport model for the high-frequency energy density evolution in bounded elastic media with general transverse or diffuse boundary conditions for the power flows. Specular-like transverse reflections and transmissions at interfaces can be treated as a particular case of the proposed model, provided that the reflection/transmission kernels have been formally identified—e.g. by the approaches cited above. Direct numerical simulations (DNS) for assemblies of beams or shells illustrate the transport regime in such systems and the onset of a diffusive regime at late times. The latter is characterized by energy equilibration rules which can be exhibited by the different schemes implemented for DNS, namely the discontinuous “Galerkin” finite element method [15] and the Monte-Carlo method [16]. The examples presented in this article underline the crucial role played by boundary and interface conditions in the emergence of a diffusive regime in piecewise homogeneous media, similarly to the effects of random heterogeneities of their densities or elastic moduli [3, 17]. These results are in basic contradiction with the vibrational conductivity analogy of the structural-acoustics literature [18, 19, 20, 21], where a diffusive regime is assumed improperly for ho-

mogeneous, unbounded media. The paper is organised as follows. First, the transport model for the evolution of the energy density in a bounded heterogeneous medium is outlined. Then two numerical examples are introduced in Sect. 3, which resumes and extends the numerical simulation of transient transport in an assembly of random thick beams [11] and presents some new simulations of multi-group transport in an assembly of thick cylindrical shells. At last Sect. 4 offers some conclusions and directions of future works.

2 Transport model

In this section one summarizes the basic results obtained in [2, 3] for the transport properties of high-frequency waves in random elastic media, and in [5] for visco-elastic materials with memory effect. The boundary conditions to be added to the evolution model outlined in Sect. 2.1 below are detailed in Sect. 2.2 for specular-like or diffusive-like reflections at outer boundaries and in Sect. 2.3 for reflections and transmissions at interfaces.

2.1 Radiative transfer in an open domain

For an heterogeneous, visco-elastic medium with statistically isotropic random perturbations of its mechanical parameters, such as the density and the Young or shear modulus, and embedded in the open domain $O \subseteq \mathbb{R}^d$ ($\mathbb{R}$ is the set of all real numbers and $d = 1, 2$ or 3), the evolution of the high-frequency energy density follows the multigroup radiative transfer equations [2, 3, 5]:

$$\partial_t w_\alpha + \mathcal{H}_\alpha w_\alpha = \sum_{\beta=1}^M \int_{S^{d-1}} \sigma_{\alpha\beta}(\mathbf{x}, |\mathbf{k}|, \hat{\mathbf{k}} \cdot \hat{\mathbf{k}}') (w_\beta(\mathbf{x}, \mathbf{k}', t) - w_\alpha(\mathbf{x}, \mathbf{k}, t)) d\mu(\hat{\mathbf{k}}'), \quad (1)$$

which couple M different energy propagation modes, or rays, for $1 \leq \alpha \leq M$. Here $\mathcal{H}_\alpha$ is the Hamiltonian associated to the eigenfrequency $\lambda_\alpha(\mathbf{x}, \mathbf{k}) = c_\alpha(\mathbf{x})|\mathbf{k}|$ of the mode α of which energy celerity is c_α , $\mathcal{H}_\alpha f \equiv \{\lambda_\alpha, f\} = \nabla_{\mathbf{k}} \lambda_\alpha \cdot \nabla_{\mathbf{x}} f - \nabla_{\mathbf{x}} \lambda_\alpha \cdot \nabla_{\mathbf{k}} f$ (the usual Poisson's bracket), S^{d-1} is the unit sphere of $\mathbb{R}^d$ with the uniform probability measure μ , and $\sigma_{\alpha\beta} \geq 0$ is the scattering cross-section which gives the rate of conversion of an energy ray β in the direction $\hat{\mathbf{k}}'$ to another ray α in the direction $\hat{\mathbf{k}}$, at position $\mathbf{x}$ and wavenumber $|\mathbf{k}|$; the standard notation $\mathbf{k} = |\mathbf{k}| \hat{\mathbf{k}}$ is used throughout this paper. In a deterministic medium all scattering cross-sections are zero and right-hand-sides vanish in Eq. (1), which reduces to Liouville transport equations. The total scattering cross-section $\Sigma_\alpha \geq 0$ for a mode α is defined by:

$$\Sigma_\alpha(\mathbf{x}, \mathbf{k}) = \sum_{\beta=1}^M \int_{S^{d-1}} \sigma_{\alpha\beta}(\mathbf{x}, |\mathbf{k}|, \hat{\mathbf{k}} \cdot \hat{\mathbf{k}}') d\mu(\hat{\mathbf{k}}'). \quad (2)$$

Here the $\sigma_{\alpha\beta}$'s are assumed to depend on the directions $\hat{\mathbf{k}}$ and $\hat{\mathbf{k}}'$ through their scalar product solely, yielding isotropic scattering, but more general scattering processes can be considered by which they depend on $\mathbf{k}$ and $\mathbf{k}'$. Note that the scalar Wigner measures $\{w_\alpha\}_{1 \leq \alpha \leq M}$, hereafter called phase-space energy densities or specific intensities, are non-negative if the initial data are non-negative. Then the space-time energy density $\mathcal{E}(\mathbf{x}, t) \in \mathbb{R}^+$ and power flow density $\boldsymbol{\Pi}(\mathbf{x}, t) \in \mathbb{R}^d$ of this medium are recovered by:

$$\mathcal{E}(\mathbf{x}, t) = \frac{1}{2} \sum_{\alpha=1}^M \int_{\mathbb{R}^d} w_\alpha(\mathbf{x}, \mathbf{k}, t) d\mathbf{k}, \quad \boldsymbol{\Pi}(\mathbf{x}, t) = \frac{1}{2} \sum_{\alpha=1}^M c_\alpha(\mathbf{x}) \int_{\mathbb{R}^d} w_\alpha(\mathbf{x}, \mathbf{k}, t) \hat{\mathbf{k}} d\mathbf{k}. \quad (3)$$

These results are adapted to elastic waves in slender structures in [8, 10, 11], but they also apply to acoustic waves, electromagnetic waves, or the Schrödinger equation in random media [2, 3, 4, 6, 7].

2.2 Radiative transfer in a bounded domain

The above system (1) is considered in the phase space $\Omega = D \times S^{d-1}$ where D is a bounded subset of O , and for $t \in [0, T]$, $T < +\infty$ but arbitrary. Wavenumber $|\mathbf{k}| > 0$ is a fixed parameter, and we shall assume as usual that the celerities $c_\alpha \in W^{1,\infty}(D)$ are positive functions with possible isolated discontinuities if D contains interfaces between different materials; see Sect. 2.3 below for further analyses. As for the scattering cross-sections, the right-hand-side collision operator in (1) is continuous on $L^2(\Omega)$ for $\sigma_{\alpha\beta} \in L^2(\Omega \times S^{d-1})$ and $\Sigma_\alpha \in L^\infty(\Omega)$, $1 \leq \alpha, \beta \leq M$. The initial conditions are known square integrable functions, $w_\alpha(\mathbf{x}, \mathbf{k}, 0) = w_\alpha^0(\mathbf{x}, \mathbf{k})$. Well-posedness of the problem is ensured by setting proper boundary conditions [22]. For example, denoting by $\hat{\mathbf{n}}$ the outward unit normal to the (smooth) boundary ∂D and introducing the boundaries $\Gamma_\pm = \{(\mathbf{x}, \hat{\mathbf{k}}) \in \partial D \times S^{d-1}; \pm \hat{\mathbf{k}} \cdot \hat{\mathbf{n}}(\mathbf{x}) > 0\}$, an incoming flux can be imposed as $c_\alpha(\mathbf{x}) w_\alpha(\mathbf{x}, \mathbf{k}, t) |\hat{\mathbf{k}} \cdot \hat{\mathbf{n}}(\mathbf{x})| = g_\alpha(\mathbf{x}, \mathbf{k}, t)$ on Γ_- for $t > 0$. Generalized boundary reflection laws can also be considered in the form:

$$\begin{aligned} & -c_\alpha(\mathbf{x}) w_\alpha(\mathbf{x}, \mathbf{k}, t) \hat{\mathbf{k}} \cdot \hat{\mathbf{n}}(\mathbf{x}) = g_\alpha(\mathbf{x}, \mathbf{k}, t) \\ & + \sum_{\beta=1}^M \int_{\hat{\mathbf{k}}' \cdot \hat{\mathbf{n}}(\mathbf{x}) > 0} \rho_{\alpha\beta}(\mathbf{x}, \hat{\mathbf{k}}, \hat{\mathbf{k}}') c_\beta(\mathbf{x}) w_\beta(\mathbf{x}, \mathbf{k}', t) \hat{\mathbf{k}}' \cdot \hat{\mathbf{n}}(\mathbf{x}) d\mu(\hat{\mathbf{k}}') \quad \text{on } \Gamma_-, \end{aligned} \quad (4)$$

$t > 0$, where $\rho_{\alpha\beta}(\mathbf{x}, \hat{\mathbf{k}}, \hat{\mathbf{k}}')$, $1 \leq \alpha, \beta \leq M$, are boundary reflectance coefficients for mode conversion from a ray β in the direction $\hat{\mathbf{k}}'$ to a ray α in the direction $\hat{\mathbf{k}}$, at $\mathbf{x} \in \partial D$. Note that these kernels depend only on the *local* geometry of the boundary (they may also depend on $|\mathbf{k}|$). They shall satisfy:

$$0 \leq \sum_{\beta=1}^M \int_{\hat{\mathbf{k}} \cdot \hat{\mathbf{n}}(\mathbf{x}) < 0} \left| \frac{c_\beta \hat{\mathbf{k}} \cdot \hat{\mathbf{n}}}{c_\alpha \hat{\mathbf{k}}' \cdot \hat{\mathbf{n}}} \right| \rho_{\beta\alpha}(\mathbf{x}, \hat{\mathbf{k}}, \hat{\mathbf{k}}') d\mu(\hat{\mathbf{k}}) \leq 1 \quad \text{a.e. } \mathbf{x} \in \partial D,$$

for the total flux reflected by the boundary for an incident ray α ; the second equality holds for a conservative boundary (total reflection). For example, specular reflection without mode conversion corresponds to $g_\alpha \equiv 0$ and:

$$\rho_{\alpha\beta}(\mathbf{x}, \hat{\mathbf{k}}, \hat{\mathbf{k}}') d\mu(\hat{\mathbf{k}}') = 2\eta(\mathbf{x}) \delta_{\alpha\beta} \otimes \delta(\hat{\mathbf{k}}' - \hat{\mathbf{k}} + 2(\hat{\mathbf{n}} \otimes \hat{\mathbf{n}})\hat{\mathbf{k}}) \quad (5)$$

with the normalization $2 \int_{\pm \hat{\mathbf{k}} \cdot \hat{\mathbf{n}}(\mathbf{x}) > 0} d\mu(\hat{\mathbf{k}}) = \int_{S^{d-1}} d\mu(\hat{\mathbf{k}}) = 1$. $\eta \leq 1$ is a boundary loss factor modeling absorption: $\eta = 1$ holds for total reflection, and $\eta < 1$ holds for partial absorption at the boundary. A common example in radiative heat transfer with $M = 1$ is the gray diffuse boundary condition for which $\eta(\mathbf{x}) = 1 - \varepsilon(\mathbf{x})$, where $\varepsilon > 0$ is the emissivity of the boundary; in that case $g(\mathbf{x}, \mathbf{k}, t)$ in Eq. (4) is the blackbody radiative flux, which is proportional to the emissivity. Another example is diffuse, or isotropic reflection (Lambertian reflectance) which corresponds to $\rho(\mathbf{x}, \hat{\mathbf{k}}, \hat{\mathbf{k}}') = 2\eta(\mathbf{x})$ independently of the observer's angle of view $\hat{\mathbf{k}}$.

2.3 Radiative transfer with a sharp interface

Assume that the physical domain D is partitioned into non-overlapping sub-domains and their interface $\gamma_D \subset \overline{D}$ is a smooth, bounded manifold of codimension 1 oriented by its unit normal $\hat{\mathbf{n}}_D$ (ignoring edges and corners at a first glance). More specifically, γ_D is any interface between these sub-domains or between a sub-domain and the exterior of D and consequently, it may also be a part or the whole of ∂D . If the celerities c_α are continuous across γ_D , the continuity of the power flow on γ_D is ensured by the conditions $c_\alpha^+ w_\alpha^+ \hat{\mathbf{k}} \cdot \hat{\mathbf{n}}_D = c_\alpha^- w_\alpha^- \hat{\mathbf{k}} \cdot \hat{\mathbf{n}}_D$ if $\hat{\mathbf{k}} \cdot \hat{\mathbf{n}}_D > 0$ (forward flux) and $c_\alpha^- w_\alpha^- \hat{\mathbf{k}} \cdot \hat{\mathbf{n}}_D = c_\alpha^+ w_\alpha^+ \hat{\mathbf{k}} \cdot \hat{\mathbf{n}}_D$ if $\hat{\mathbf{k}} \cdot \hat{\mathbf{n}}_D < 0$ (backward flux), where $w_\alpha^\pm(\mathbf{x}, \mathbf{k}, t) = \lim_{h \rightarrow 0^+} w_\alpha(\mathbf{x} \pm h\hat{\mathbf{n}}_D, \mathbf{k}, t)$ for a.e. $\mathbf{x} \in \gamma_D$, and $w_\alpha^+(\mathbf{x}, \mathbf{k}, t) = 0$ whenever $\mathbf{x} \in \partial D$; likewise, $c_\alpha^\pm(\mathbf{x}) = \lim_{h \rightarrow 0^+} c_\alpha(\mathbf{x} \pm h\hat{\mathbf{n}}_D)$ a.e. $\mathbf{x} \in \gamma_D$. If however the celerities are discontinuous across γ_D , such that $c_\alpha^+ \neq c_\alpha^-$, these expressions may be generalized to the form:

$$\begin{aligned} c_\alpha^+(\mathbf{x}) w_\alpha^+(\mathbf{x}, \mathbf{k}, t) \hat{\mathbf{k}} \cdot \hat{\mathbf{n}}_D(\mathbf{x}) = \\ \sum_{\beta=1}^M \left(\int_{\hat{\mathbf{k}}' \cdot \hat{\mathbf{n}}_D(\mathbf{x}) > 0} \tau_{\alpha\beta}^-(\mathbf{x}, \hat{\mathbf{k}}, \hat{\mathbf{k}}') c_\beta^-(\mathbf{x}) w_\beta^-(\mathbf{x}, \mathbf{k}', t) \hat{\mathbf{k}}' \cdot \hat{\mathbf{n}}_D(\mathbf{x}) d\mu(\hat{\mathbf{k}}') \right. \\ \left. - \int_{\hat{\mathbf{k}}' \cdot \hat{\mathbf{n}}_D(\mathbf{x}) < 0} \rho_{\alpha\beta}^+(\mathbf{x}, \hat{\mathbf{k}}, \hat{\mathbf{k}}') c_\beta^+(\mathbf{x}) w_\beta^+(\mathbf{x}, \mathbf{k}', t) \hat{\mathbf{k}}' \cdot \hat{\mathbf{n}}_D(\mathbf{x}) d\mu(\hat{\mathbf{k}}') \right) \quad (6) \end{aligned}$$

if $\hat{\mathbf{k}} \cdot \hat{\mathbf{n}}_D(\mathbf{x}) > 0$, and:

$$\begin{aligned}
c_{\alpha}^{-}(\mathbf{x})w_{\alpha}^{-}(\mathbf{x}, \mathbf{k}, t)\hat{\mathbf{k}} \cdot \hat{\mathbf{n}}_D(\mathbf{x}) = \\
\sum_{\beta=1}^M \left(\int_{\hat{\mathbf{k}}' \cdot \hat{\mathbf{n}}_D(\mathbf{x}) < 0} \tau_{\alpha\beta}^{+}(\mathbf{x}, \hat{\mathbf{k}}, \hat{\mathbf{k}}') c_{\beta}^{+}(\mathbf{x}) w_{\beta}^{+}(\mathbf{x}, \mathbf{k}', t) \hat{\mathbf{k}}' \cdot \hat{\mathbf{n}}_D(\mathbf{x}) d\mu(\hat{\mathbf{k}}') \right. \\
\left. - \int_{\hat{\mathbf{k}}' \cdot \hat{\mathbf{n}}_D(\mathbf{x}) > 0} \rho_{\alpha\beta}^{-}(\mathbf{x}, \hat{\mathbf{k}}, \hat{\mathbf{k}}') c_{\beta}^{-}(\mathbf{x}) w_{\beta}^{-}(\mathbf{x}, \mathbf{k}', t) \hat{\mathbf{k}}' \cdot \hat{\mathbf{n}}_D(\mathbf{x}) d\mu(\hat{\mathbf{k}}') \right) \quad (7)
\end{aligned}$$

if $\hat{\mathbf{k}} \cdot \hat{\mathbf{n}}_D(\mathbf{x}) < 0$. In the above equations $\rho_{\alpha\beta}^{\pm}$ and $\tau_{\alpha\beta}^{\pm}$ are the left (−) and right (+) reflectance/transmittance coefficients, respectively, of the interface γ_D where the c_{α} 's may be discontinuous. They satisfy the conservation rules:

$$\begin{aligned}
\sum_{\beta=1}^M \left(\int_{\pm \hat{\mathbf{k}} \cdot \hat{\mathbf{n}}_D(\mathbf{x}) < 0} \int_{\pm \hat{\mathbf{k}}' \cdot \hat{\mathbf{n}}_D(\mathbf{x}) < 0} \tau_{\beta\alpha}^{\pm}(\mathbf{x}, \hat{\mathbf{k}}, \hat{\mathbf{k}}') v(\hat{\mathbf{k}}') (\hat{\mathbf{k}}' \cdot \hat{\mathbf{n}}_D(\mathbf{x})) d\mu(\hat{\mathbf{k}}') d\mu(\hat{\mathbf{k}}) \right. \\
\left. - \int_{\pm \hat{\mathbf{k}} \cdot \hat{\mathbf{n}}_D(\mathbf{x}) > 0} \int_{\pm \hat{\mathbf{k}}' \cdot \hat{\mathbf{n}}_D(\mathbf{x}) < 0} \rho_{\beta\alpha}^{\pm}(\mathbf{x}, \hat{\mathbf{k}}, \hat{\mathbf{k}}') v(\hat{\mathbf{k}}') (\hat{\mathbf{k}}' \cdot \hat{\mathbf{n}}_D(\mathbf{x})) d\mu(\hat{\mathbf{k}}') d\mu(\hat{\mathbf{k}}) \right) \\
\leq \int_{\pm \hat{\mathbf{k}} \cdot \hat{\mathbf{n}}_D(\mathbf{x}) < 0} v(\hat{\mathbf{k}}) (\hat{\mathbf{k}} \cdot \hat{\mathbf{n}}_D(\mathbf{x})) d\mu(\hat{\mathbf{k}}) \quad \text{a.e. } \mathbf{x} \in \gamma_D
\end{aligned}$$

for, say, any $v \in L^1(S^{d-1}, |\hat{\mathbf{k}} \cdot \hat{\mathbf{n}}| d\mu)$, stating that the total flux impinging the boundary is either reflected or transmitted with or without losses; a strict inequality holds in the former case, while a strict equality holds in the latter case. Note that these conditions relate transmittances $\tau_{\alpha\beta}^{\pm}$ and reflectances $\rho_{\alpha\beta}^{\pm}$ by the requirement that the normal energy flow $\mathbf{\Pi} \cdot \hat{\mathbf{n}}_D$ be conserved across the interface; it turns out that the energy density itself is not necessarily continuous across that interface.

Example 1. Simply choosing $\rho_{\alpha\beta}^{\pm} \equiv 0$ and $\tau_{\alpha\beta}^{\pm} \equiv \delta(c_{\alpha}^{\mp} \hat{\mathbf{k}} - c_{\beta}^{\pm} \hat{\mathbf{k}}')$ (no scattering on γ_D) yields a continuous upwind flux, as above for continuous celerities. $\square$

Example 2. If γ_D is a part of the boundary ∂D with $\tau_{\alpha\beta}^{-} = 0$ and no incoming flux, $w_{\beta}^{+} = 0$ for $1 \leq \beta \leq M$, the boundary flow of Eq. (4) with $g_{\alpha} \equiv 0$ is directly recovered. $\square$

Example 3. For $d \geq 2$ and considering specular-like transverse reflection and transmission of polarized elastic waves on a sharp interface, the reflectance and transmittance coefficients $\rho_{\alpha\beta}^{\pm}$ and $\tau_{\alpha\beta}^{\pm}$ have the form:

$$\begin{aligned}
\rho_{\alpha\beta}^{\pm}(\mathbf{x}, \hat{\mathbf{k}}, \hat{\mathbf{k}}') d\mu(\hat{\mathbf{k}}) &= \mathcal{R}_{\alpha\beta}^{\pm}(\mathbf{x}, \hat{\mathbf{k}}') \delta(\hat{\mathbf{k}} - \hat{\mathbf{R}}_{\alpha\beta}^{\pm}(\hat{\mathbf{k}}')) , \\
\tau_{\alpha\beta}^{\pm}(\mathbf{x}, \hat{\mathbf{k}}, \hat{\mathbf{k}}') d\mu(\hat{\mathbf{k}}) &= \mathcal{T}_{\alpha\beta}^{\pm}(\mathbf{x}, \hat{\mathbf{k}}') \delta(\hat{\mathbf{k}} - \hat{\mathbf{T}}_{\alpha\beta}^{\pm}(\hat{\mathbf{k}}')) ,
\end{aligned} \quad (8)$$

where $\hat{\mathbf{R}}_{\alpha\beta}^{\pm}$ and $\hat{\mathbf{T}}_{\alpha\beta}^{\pm}$ are one-to-one mappings from $\omega_D^{\mp}(\mathbf{x})$ into $\omega_D^{\pm}(\mathbf{x})$ and $\omega_D^{\mp}(\mathbf{x})$, respectively, with $\omega_D^{\pm}(\mathbf{x}) = \{\hat{\mathbf{k}} \in S^{d-1}; \pm \hat{\mathbf{k}} \cdot \hat{\mathbf{n}}_D(\mathbf{x}) > 0\}$. More precisely, they are defined on the open *hyperbolic subsets* $\mathcal{C}^{\pm}(\mathbf{x}, \Theta)$ of $\omega_D^{\pm}(\mathbf{x})$, which are the intersections with S^{d-1} of the cones of vertex $\mathbf{x}$, axis $\pm \hat{\mathbf{n}}_D(\mathbf{x})$, and solid angles $\Omega_d \sin^2(\Theta/2)$, where Θ are critical angles arising in conjunction with Descartes law

stating that the tangential wave vector remains constant in the reflection and transmission process at a point $\mathbf{x}$ on the interface γ_D . They are such that $\Theta \equiv \theta_{\alpha\beta}^{--}(\mathbf{x})$ for $\hat{\mathbf{R}}_{\alpha\beta}^-$, $\Theta \equiv \theta_{\alpha\beta}^{+-}(\mathbf{x})$ for $\hat{\mathbf{T}}_{\alpha\beta}^-$, $\Theta \equiv \theta_{\alpha\beta}^{++}(\mathbf{x})$ for $\hat{\mathbf{R}}_{\alpha\beta}^+$, and $\Theta \equiv \theta_{\alpha\beta}^{-+}(\mathbf{x})$ for $\hat{\mathbf{T}}_{\alpha\beta}^+$, where $\theta_{\alpha\beta}^{\pm\pm}(\mathbf{x}) = \arcsin(c_{\beta}^{\pm}/c_{\alpha}^{\pm})$. Let $T_{\mathbf{x}}\gamma_D$ be the tangent space to γ_D at $\mathbf{x}$ and let $P_{\mathbf{x}} = \mathbb{I}_d - \hat{\mathbf{n}}_D(\mathbf{x}) \otimes \hat{\mathbf{n}}_D(\mathbf{x})$ be the orthogonal projection from $\mathbb{R}^d$ into that space, where $\mathbb{I}_d$ stands for the identity matrix in $\mathbb{R}^d$; then Descartes law writes:

$$\frac{c_{\beta}^{\pm}}{c_{\alpha}^{\pm}} P_{\mathbf{x}} \hat{\mathbf{R}}_{\alpha\beta}^{\pm}(\hat{\mathbf{k}}) = \frac{c_{\beta}^{\pm}}{c_{\alpha}^{\mp}} P_{\mathbf{x}} \hat{\mathbf{T}}_{\alpha\beta}^{\pm}(\hat{\mathbf{k}}) = P_{\mathbf{x}} \hat{\mathbf{k}}, \quad \forall \hat{\mathbf{k}} \in S^{d-1}, \quad \forall \alpha, \beta = 1, 2 \dots M.$$

In addition, the mappings $\hat{\mathbf{R}}_{\alpha\beta}^{\pm}$ and $\hat{\mathbf{T}}_{\alpha\beta}^{\pm}$ have the properties (reciprocity):

$$-\hat{\mathbf{R}}_{\beta\alpha}^{\pm}(-\hat{\mathbf{R}}_{\alpha\beta}^{\pm}(\hat{\mathbf{k}})) = \hat{\mathbf{k}}, \quad \hat{\mathbf{k}} \in \mathcal{C}^{\mp}(\mathbf{x}, \theta_{\alpha\beta}^{\pm\pm}), \quad -\hat{\mathbf{T}}_{\beta\alpha}^{\mp}(-\hat{\mathbf{T}}_{\alpha\beta}^{\pm}(\hat{\mathbf{k}})) = \hat{\mathbf{k}}, \quad \hat{\mathbf{k}} \in \mathcal{C}^{\mp}(\mathbf{x}, \theta_{\alpha\beta}^{\mp\pm}),$$

and (symmetric reflection without mode conversion — see also Eq. (5)):

$$\hat{\mathbf{R}}_{\alpha\alpha}^{\pm}(\hat{\mathbf{k}}) = (P_{\mathbf{x}} - \hat{\mathbf{n}}_D \otimes \hat{\mathbf{n}}_D) \hat{\mathbf{k}}, \quad \hat{\mathbf{k}} \in \mathcal{C}^{\mp}(\mathbf{x}, \theta_{\alpha\alpha}^{\pm\pm}) \equiv \omega_D^{\mp}(\mathbf{x}).$$

On the *elliptic sets* $\omega_D^{\pm}(\mathbf{x}) \setminus \overline{\mathcal{C}^{\pm}(\mathbf{x}, \Theta)}$, $\alpha \neq \beta$, one simply has $\mathcal{R}_{\alpha\beta}^{\pm} = \mathcal{T}_{\alpha\beta}^{\pm} = 0$. The *glancing sets* are such that $\hat{\mathbf{R}}_{\alpha\beta}^{\pm}(\hat{\mathbf{k}}) \cdot \hat{\mathbf{n}}_D = \hat{\mathbf{T}}_{\alpha\beta}^{\pm}(\hat{\mathbf{k}}) \cdot \hat{\mathbf{n}}_D = 0$ but they are ignored in our formulation because a theoretical model lacks at present to describe them, especially the corresponding kernels $\mathcal{R}_{\alpha\beta}^{\pm}$ and $\mathcal{T}_{\alpha\beta}^{\pm}$ in the elastic case. The latter are otherwise positive functions and, at least, smooth on $\mathcal{C}^{\mp}(\mathbf{x}, \Theta)$. A rigorous derivation of the reflection/transmission operators (8) for elastic media with Dirichlet or Neumann boundary conditions away from the glancing set is proposed in [13], while a formal derivation for slender elastic structures is proposed in [11]. Both are adapted from the ideas developed by Miller for scalar waves in [14] (this work also analyzes rigorously the glancing set). This case has also been studied numerically by Jin & Yin [23]. $\square$

3 Numerical examples

In Sect. 3.1 the example of [11] of scalar transport in coupled Timoshenko beams is resumed and extended to random heterogeneities. Sect. 3.2 deals with two coupled thick cylindrical shells. In these examples the radiative transfer equations (1) with specular-like boundary and interface (at the junctions) conditions are solved numerically by two distinct schemes: a Runge-Kutta “Galerkin” discontinuous finite element method is used for the first example, while a fully vectorized direct Monte-Carlo method is used for the second example.

3.1 Coupled beams

We first consider an assembly of two random Timoshenko beams $] -L, 0[\cup] 0, L[, L > 0$, with a junction at $x = 0$ such that beam #2 forms an angle ϕ with beam #1. Three types of energy rays propagate in such a system: a pure shear transverse mode at the celerities c_{Tj} ($j = 1$ or 2 for either beam #1 or beam #2) denoted by subscript T, a pure bending mode at the celerities c_{Pj} denoted by subscript Pb, and a longitudinal mode at the celerities c_{Pj} denoted by subscript Pn; see [10, 11] for some details on the derivation. Therefore the energy density evolution in this system is depicted by three coupled, one-dimensional radiative transfer equations ($M = 3$ and $d = 1$ such that $\hat{k} \equiv \text{sign}(k)$) written as:

$$\frac{1}{c_\alpha} \partial_t w_\alpha + \hat{k} \partial_x w_\alpha + \eta w_\alpha = \Sigma_\alpha(|k|)(w_\alpha(-k) - w_\alpha(k)), \quad \alpha = \text{T, Pb, Pn}, \quad (9)$$

where the coupling is ensured by the left $(-)$ and right $(+)$ power reflection and transmission kernels $\mathcal{R}_{\alpha\beta}^\pm(\phi)$, $\mathcal{T}_{\alpha\beta}^\pm(\phi)$ of the junction. Indeed, for thick beams no coupling of modes occurs through the collision operator on the right-hand-side in Eq. (1) [10, 11]. The expressions of $\mathcal{R}_{\alpha\beta}^\pm$ and $\mathcal{T}_{\alpha\beta}^\pm$ as functions of ϕ and beams mechanical and geometrical characteristics are derived in [11], using a high-frequency model compatible with the propagation characteristics obtained by the transport theory. This study has also shown that the power reflection coefficients at a beam end for either Dirichlet (clamped) or Neumann (free) boundary condition were all equal to 1, without mode conversion of any type; therefore specular reflections are considered at the boundaries $x = \pm L$. The total scattering cross-sections are (Gaussian model) $\Sigma_\alpha(k_0) = \frac{\sqrt{\pi}}{2} r_\alpha k_0^2 e^{-k_0^2}$, $k_0 = \ell k$, where ℓ is a correlation length of the parameters random fluctuations and r_α is a scaling factor such that $|r_\alpha| \leq 1$. All computations are performed for the initial conditions:

$$w_{\text{T}}^0(x, k) = e^{-\ln 2 \left(\frac{x-x_0}{r_0} \right)^2} \otimes \delta(\hat{k} - \varepsilon_0), \quad w_{\text{Pb}}^0 = w_{\text{Pn}}^0 \equiv 0,$$

with $x_0 = -0.1 \times L$, $r_0 = 0.4 \times L$, and $\varepsilon_0 = +1$. Beam celerities are such that $c_{\text{T2}}/c_{\text{T1}} = c_{\text{P2}}/c_{\text{P1}} = \sqrt{2}$, their Poisson's ratio being $\nu_1 = \nu_2 = 0.3$, and the junction angle is $\phi = \pi/3$.

The coupled radiative transfer equations (9) are solved numerically by the Runge-Kutta ‘‘Galerkin’’ discontinuous finite element method introduced in [25] (see also [15], for example, for a general reference). In the proposed scheme, the reflectance/transmittance coefficients are directly taken into account by the numerical fluxes. In this example, the assembly is partitioned into 100 uniform spatial elements. Legendre polynomials up to the fourth order are used as local basis functions on them, and Eqs. (9) are solved in time by a second-order SSP (Strong-Stability-Preserving) Runge-Kutta scheme [24]. The Courant number for the simulations is fixed at $\nu \equiv c_{\text{T2}} \frac{\Delta t}{\Delta x} = 2.5 \cdot 10^{-2}$. Fig. 1 and Fig. 2 display our results for either $r_\alpha = 0$ or $r_\alpha = 0.3$ and $k_0 = 1.5$, respectively, for the scattering cross-sections. The first case corresponds to pure transport without scattering in a deterministic, piecewise

homogeneous assembly, and the second case corresponds to full radiative transfer in a randomly heterogeneous assembly. The time scale is $T = L/c_{T2}$, the time needed by the wave front to reach the extremity $x = L$ of the homogeneous assembly starting from the junction. The thin vertical line indicates the location of the junction. The wavefronts of shear ($\alpha = T$) and bending/axial ($\alpha = P$) energy densities propagating at the speeds c_{Tj} and c_{Pj} , respectively, are marked by small capital letters. A small amount of damping $\eta = 0.02 \text{ m}^{-1}$ has been added such that the total energy within the system at a fixed wavenumber $|k|$ evolves as (see Eq. (3)):

$$\mathcal{E}(t) = \sum_{\alpha=T,Pb,Pn} e^{-\eta c_{\alpha} t} \mathcal{E}_{\alpha}^0, \quad \mathcal{E}_{\alpha}^0 = \int_{-L}^L w_{\alpha}^0(x, \pm|k|) dx. \quad (10)$$

This exact result is recovered numerically by our scheme, as shown by Fig. 3 and Fig. 4. They display the overall mechanical energies $\mathcal{E}_j^h(t)$ computed within each beam by $\mathcal{E}_1^h(t) = \int_{-L}^0 \mathcal{E}^h(x, t) dx$, $\mathcal{E}_2^h(t) = \int_0^L \mathcal{E}^h(x, t) dx$, and their sum, normalized by the total energy introduced into the system $\mathcal{E}(0) = \sum_{\alpha} \mathcal{E}_{\alpha}^0 = \mathcal{E}_T^0$. The computed and exact total energies perfectly match on these plots: one concludes that the proposed scheme in dimension one is numerically conservative.

These results exhibit the following features: one observes a beating phenomenon of the total energy within each subsystem, by which it is transferred back and forth from the first beam to the second one and the other way round. At late time an equipartition regime holds between both beams. It is described by diffusion equations as explained in [17] for example. It should be noted that this diffusive regime also arises in the case of a piecewise homogeneous assembly (see Fig. 3), the onset of diffusion being prompted by the mixing of wave fronts and modes at the junction, and to a lesser extent at the boundaries. This mixing process is accelerated by the presence of random heterogeneities, as seen on Fig. 4.

Fig. 1: Evolution of the energy density in an assembly of two homogeneous Timoshenko beams with $\phi = \frac{\pi}{3}$, $c_{T2}/c_{T1} = c_{P2}/c_{P1} = \sqrt{2}$, $v_1 = v_2 = 0.3$, $\eta = 0.02 \text{ m}^{-1}$, and $T = L/c_{T2}$

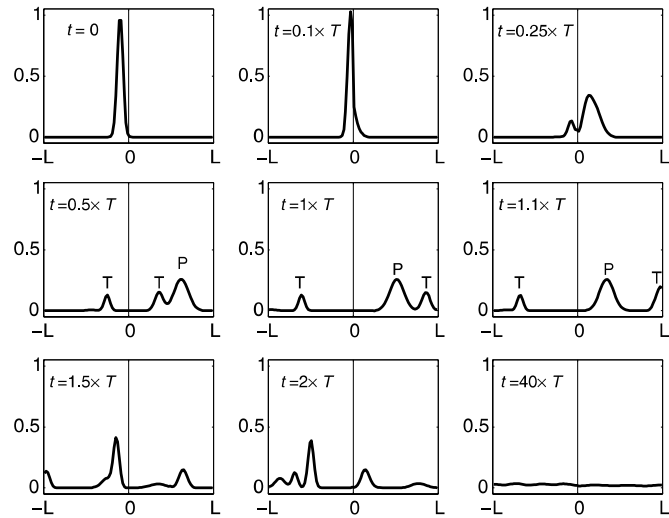


Fig. 2: Evolution of the energy density in an assembly of randomly heterogeneous Timoshenko beams with $\phi = \frac{\pi}{3}$, $c_{T2}/c_{T1} = c_{P2}/c_{P1} = \sqrt{2}$, $v_1 = v_2 = 0.3$, $\eta = 0.02 \text{ m}^{-1}$, $T = L/c_{T2}$, and $r_\alpha = 0.3$, $k_0 = 1.5$ for the scattering cross-sections

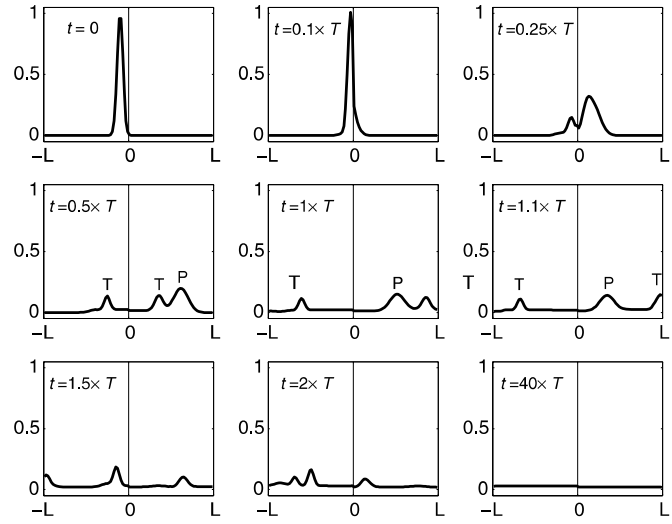


Fig. 3: Evolution of the total vibrational energies in an assembly of two homogeneous Timoshenko beams with $\phi = \frac{\pi}{3}$, $c_{T2}/c_{T1} = c_{P2}/c_{P1} = \sqrt{2}$, $v_1 = v_2 = 0.3$, $\eta = 0.02 \text{ m}^{-1}$, and $T = L/c_{T2}$. Thick solid line: beam #1, thick dashed line: beam #2, thin solid line: computed total energy, thin dashed line: exact total energy

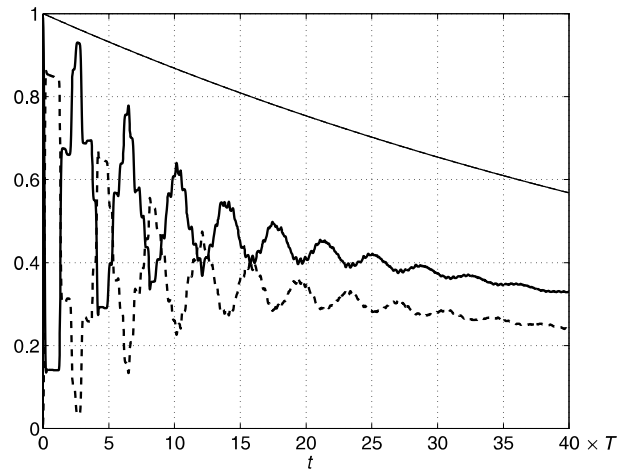
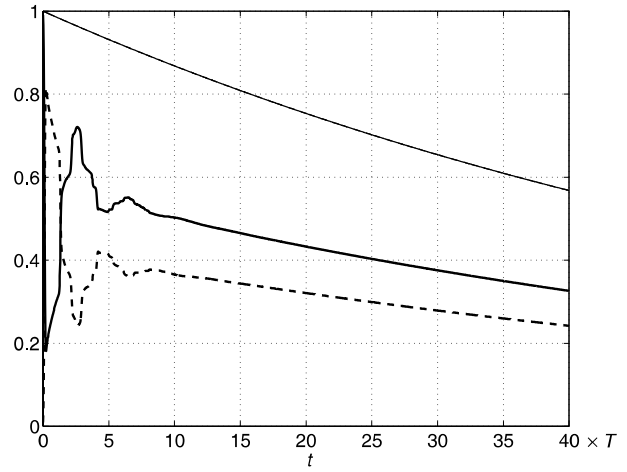


Fig. 4: Evolution of the total vibrational energies in an assembly of two randomly heterogeneous Timoshenko beams with $\phi = \frac{\pi}{3}$, $c_{T2}/c_{T1} = c_{P2}/c_{P1} = \sqrt{2}$, $v_1 = v_2 = 0.3$, $\eta = 0.02 \text{ m}^{-1}$, $T = L/c_{T2}$, and $r_\alpha = 0.3$, $k_0 = 1.5$ for the scattering cross-sections. Thick solid line: beam #1, thick dashed line: beam #2, thin solid line: computed total energy, thin dashed line: exact total energy



3.2 Coupled shells

Next we consider an assembly of two thick cylindrical shells, the first one with a constant radius and the second one with a variable radius. Both cylinders have the same length L which is also twice the constant radius of the first shell. The angle

between the shells is again denoted by ϕ , and their mean surfaces are denoted by D_1 and D_2 , respectively. Five types of energy rays propagate in such a system: a pure shear transverse mode at the celerities c_{Tj} ($j = 1$ or 2 for either shell #1 or shell #2) denoted by subscript T, and coupled longitudinal and transverse modes P and S each of multiplicity 2 corresponding to either in-plane (subscript n) or bending energies (subscript b), at the celerities c_{Pj} and c_{Sj} respectively; see [8] for some details on the derivation. Then the energy density evolution in this system is depicted by five coupled radiative transfer equations (1) with $M = 5$ —since $\alpha = T, Pb, Pn, Sb$ or Sn —and $d = 2$, where the coupling is ensured by the left (−) and right (+) power reflection and transmission kernels $\mathcal{R}_{\alpha\beta}^{\pm}(\phi, \mathbf{k}_t)$, $\mathcal{T}_{\alpha\beta}^{\pm}(\phi, \mathbf{k}_t)$ of the shell junction. Their expressions as functions of ϕ , the incident tangent wave vector $\mathbf{k}_t = P_{\mathbf{x}_t} \mathbf{k}$ on the junction line $\gamma_D \ni \mathbf{x}_t$, and shells mechanical and geometrical characteristics are derived in [11], using a high-frequency model compatible with the propagation characteristics obtained by the transport theory within plates. As the curvature tensor has no influence on the high-frequency regime in this model, the same reflection/transmission kernels as for flat plates are used in this example. Neumann boundary conditions are considered at the extremities of the shells, and the corresponding power reflection coefficients have been derived in [11] as well. Here it has been shown that the shear mode is uncoupled to the in-plane and bending modes by boundary reflections, however the longitudinal P and transverse S modes are coupled. Otherwise all these modes are fully coupled by the junction. The initial condition is an isotropic shear impact $w_T^0(\mathbf{x}, \mathbf{k}) = \delta(\mathbf{x} - \mathbf{x}_0)$, independently of $\mathbf{k}$, all other modes being initially unloaded. Here $\mathbf{x}_0 = (-L/10, 0)$ and the junction line is the circle $\gamma_D = \{\mathbf{x} = (0, y), y \in [0, \pi L]\}$ in curvilinear coordinates. The Young's moduli, densities, and thicknesses of both shells are equal, their Poisson's ratios are $\nu_1 = 0.1$ and $\nu_2 = 0.4$, and the junction angle is $\phi = \pi/12$.

The coupled radiative transfer equations (1) are solved in this case by a direct Monte-Carlo method [16]. The computer code developed for this application has been fully vectorized, in such a way that numerical simulations of $O(10^6)$ ray paths with $O(10^2)$ time steps can be performed in less than one hour of CPU time on a personal computer. Fig. 5 through Fig. 7 below display our results for pure transport without scattering in a deterministic, piecewise homogeneous assembly for illustration purposes. Propagation in random shells was considered in [8]. The time scale is $T = L/c_{T2}$, and a small amount of damping $\eta = 0.02 \text{ m}^{-1}$ has been added as for the beam assembly of the first example. Fig. 5 displays the evolution of the energy density within the shell assembly, while Fig. 6 displays the evolution of the overall mechanical energies within each shell and their sum. In addition Fig. 7 displays the ratios $\mathcal{E}_{Pj}^h(t)/\mathcal{E}_{Sj}^h(t)$ of the total longitudinal and transverse energies within each shells, respectively, for the same data as above except $\eta = 0$, where:

$$\mathcal{E}_{Pj}^h(t) = \sum_{\alpha=Pn, Pb} \int_{D_j \times S^1} w_{\alpha}^h(\mathbf{x}, \mathbf{k}, t) d\mathbf{x} d\mu(\hat{\mathbf{k}}), \quad j = 1, 2,$$

and a similar expression for $\mathcal{E}_{Sj}^h(t)$. For an isotropic collision operator, their diffusive limit at large times is proved to be the ratio of the celerities c_{Sj}/c_{Pj} to the power of

the physical dimension, $d = 2$ in the present case, independently of the source type and scatterers [3, 17]. Here this universal equilibration rule is apparently recovered for anisotropic scatterers constituted by the junction and the boundaries. Again, it can be concluded that the numerical scheme is conservative, and that it exhibits the equilibration rule of energies between both shells in the late-time diffusive regime. The emergence of the latter is prompted here by the sole contribution of the energy rays numerous reflections and transmissions at the boundaries and the junction, together with mode conversions. A quite unexpected observation is the promptness of the stabilization of the energy ratios with time. This result, however, ignores the contribution of the glancing rays possibly trapped by the boundaries and the junction line, because a theoretical model lacks at present to describe them. Indeed it is known from experiments that the vibrational energy is very much likely to be guided by the interfaces, stiffeners or junctions in a real-life complex structure, so these effects should not be disregarded. Further developments are needed on this issue.

Fig. 5: Evolution of the energy density in an assembly of two homogeneous cylindrical shells with $\phi = \frac{\pi}{12}$, $v_1 = 0.1$, $v_2 = 0.4$, $\eta = 0.02 \text{ m}^{-1}$, and $T = L/c_{T2}$

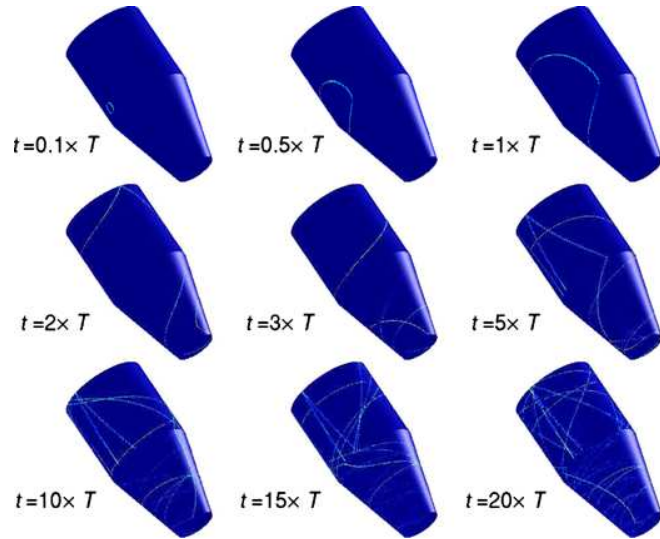


Fig. 6: Evolution of the total vibrational energies within each shell of the assembly with $\phi = \frac{\pi}{12}$, $v_1 = 0.1$, $v_2 = 0.4$, $\eta = 0.02 \text{ m}^{-1}$, and $T = L/c_{T2}$. Thick solid line: shell #1, thick dashed line: shell #2, thin solid line: computed total energy

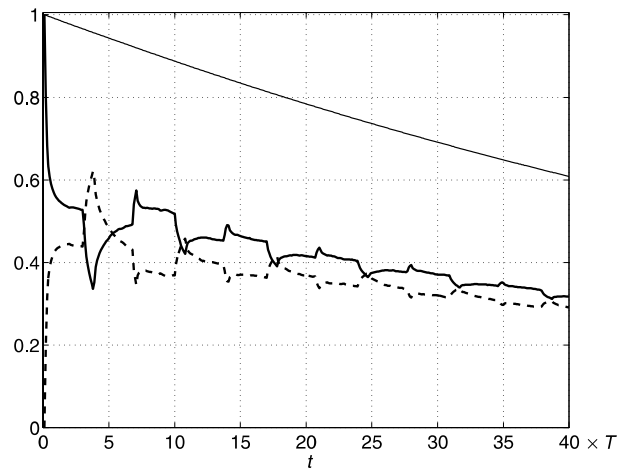
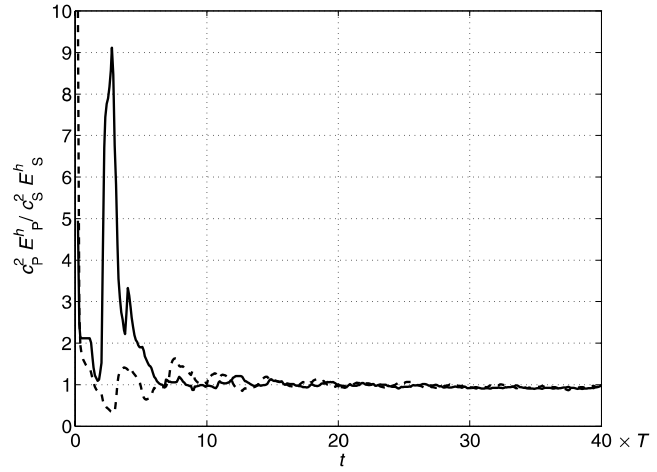


Fig. 7: Evolution of the ratio $c_{Pj}^2 \mathcal{E}_{Pj}^h(t) / c_{Sj}^2 \mathcal{E}_{Sj}^h(t)$ of the total longitudinal and transverse energies within each shell of the assembly with $\phi = \frac{\pi}{12}$, $v_1 = 0.1$, $v_2 = 0.4$, $\eta = 0$, and $T = L/c_{T2}$. Solid line: shell #1 ($j = 1$), dashed line: shell #2 ($j = 2$)



4 Conclusions

A transport model for the elastic energy propagation in bounded heterogeneous media has been proposed. It accounts for general transverse reflection/transmission effects at boundaries and interfaces, including specular-like reflections according to Descartes law as a particular case. This model is amenable to be solved numerically by the discontinuous finite element method or the Monte-Carlo method, among other possible schemes. Both methods have been implemented and tested on examples of assemblies of thick beams or cylindrical shells. DNS illustrate the transport regime for these systems, and the onset of a diffusive regime at late times can also be exhibited. It is characterized by energy equilibration rules very much comparable to the assumption of modal equipartition invoked in the statistical energy analysis (SEA) of structural-acoustics systems [26]. However a rigorous mathematical model of the diffusive regime for bounded media lacks at present. It should be able to predict the equilibration ratios as function of the sub-systems mechanical parameters, as well as their unexpectedly rapid stabilization with time in two dimensions. Another (still) open issue is the consideration of the glancing set of energy rays — the diffractive and gliding ones — for the part of the power flows possibly trapped by the interfaces and junctions. Both aspects are the subject of ongoing research.

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